



A COMMUNITY OF CHOICE

CITY ENGINEER'S OFFICE
2101 O'Neil Avenue
Cheyenne, Wyoming 82001

MEMO

TO: Honorable Mayor Patrick J. Collins, City of Cheyenne
Members of the City Council, City of Cheyenne

FROM: Sam Berta, Staff Engineer

DATE: April 10, 2023

SUBJECT: Professional Service Agreement between the City of Cheyenne and AVI P.C., for the design of the Dry Creek Dredging and Maintenance Project.

Please find the attached Professional Services Agreement between the City and AVI P.C. for engineering services related to the design of the Dry Creek Dredging and Maintenance Project. This agreement will provide the design for maintenance of Dry Creek and restoring it to normal conditions. This contract will be a 2-year contract providing 2 sets of project documents and assisting with future planning for continued maintenance. This project will provide the needed services to help the City address the crucial task of maintaining our drainage systems.

Engineering Staff evaluates fee proposals for engineering services based on time and material estimates for the specific project. Those costs are readily tracked and verified during design operations and the City only pays for the actual time expended. We believe this agreement will provide the required services to complete this project.

The fees include services for project management, drainage design, topographical surveying, plan creation and planning services. Funding for this project will come from the 2019 - 2022 1% Specific Purpose Tax Fund (5th Penny). The total professional services will be a cost not-to-exceed \$199,601.50. We respectfully request your approval of this professional services agreement.

City Contract No. _____

PROFESSIONAL SERVICES AGREEMENT
for
**Dry Creek Dredging & Maintenance Project Design, and Bidding /
Construction Services**

THIS AGREEMENT is made and entered into by and between the City of Cheyenne, located at 2101 O'Neil Avenue, Cheyenne, Wyoming 82001 ("City"), and AVI Professional Corporation, located at 1103 Old Town Lane, Suite 101, Cheyenne, Wyoming 82009 ("Consultant"). This Agreement shall become effective upon the date of the last signature affixed hereto.

RECITALS

In consideration of the mutual covenants and obligations herein expressed, it is agreed by and between the parties as follows:

1. SCOPE OF WORK

The Consultant agrees to provide services as described for the above-mentioned project, in accordance with the Scope of Work attached hereto as Exhibit A, incorporated by reference and expressly made a part hereof.

2. TIME OF PERFORMANCE

The work included in this Scope of Work will be completed by **May 1, 2025**.

3. CONSULTANT RESPONSIBILITY

In providing services under this Agreement, the Consultant shall perform in a manner consistent with that degree of care and skill ordinarily exercised by members of the same profession currently practicing under similar circumstances. Upon notice to the Consultant, the Consultant will, without additional compensation, correct those services not meeting such a standard.

4. COMPLIANCE WITH LAWS

The Parties agree that they will perform their obligations as provided in this Agreement in accordance with all applicable laws and ordinances.

5. DRUG-FREE WORKPLACE

In compliance with the Drug Free Workplace Act of November 1988, the City of Cheyenne has established an Alcohol and Controlled Substance Policy that pertains to alcohol and drug usage by City Employees. All architects, engineers and other consultants under contract with City of Cheyenne, and their employees and sub-consultants, are required to comply with the provisions of the City's Alcohol and Controlled Substance Policy for drug

and/or alcohol usage on City property or other sites occupied by the Consultant while performing the duties and responsibilities of the Agreement. It is the responsibility of the Consultant to familiarize themselves with the requirements of this policy and to inform all their employees and sub-consultants of the requirements and ensure their compliance. If the Consultant, their employees or sub-consultants are found in violation of this policy, the Agreement may be terminated.

6. INDEMNIFICATION/HOLD HARMLESS

The Consultant agrees to indemnify, hold harmless and defend the City from and against any and all liabilities, claims, penalties, forfeitures and suits, and the cost and expenses incident thereto, including reasonable attorney's fees, which may hereafter arise as a result the performance of the Consultant's duties, including death or bodily injury to any person, destruction or damage to any property, contamination of or adverse effects on the environment, or any violation of governmental laws, regulations or orders to the extent caused by (1) the Consultant's breach of any term or provision of this Agreement; or (2) any negligent or wrongful act, error or omission by the Consultant, or its employees or sub-consultants in the performance of this Agreement. The Consultant acknowledges that it may incur a financial obligation to the City pursuant to the terms of this paragraph.

7. INSURANCE REQUIREMENTS

The Consultant must provide proof of the following insurance coverages:

Commercial General Liability Insurance

For claims arising out of bodily injury, illness or death, or from damage to or destruction of property of others, including loss or use thereof, with minimum limits of \$1,000,000 per occurrence, \$2,000,000 aggregate for the entire term of the Agreement.

Workers' Compensation

Workers' compensation coverage shall be in effect for the entire term of the Agreement, as required by Wyoming law, for all employees or agents providing services under this Agreement. Consultant shall provide the City with proof of workers' compensation or employer's liability insurance coverage.

Professional Liability Insurance

The Consultant shall provide proof of professional liability insurance or errors and omissions liability insurance in an amount not less than \$500,000 to protect the City from any and all claims arising from the Consultant's negligence in the performance of duties under this Agreement. The City prefers that this liability insurance coverage be provided pursuant to an "occurrence" policy.

If this coverage is provided pursuant to a "claims made" policy:

- a. Consultant shall, concurrently with the execution of this Agreement, provide the City with a certificate of insurance demonstrating that such

- coverage is or shall be in effect at the time the Consultant begins the provision of services under this Agreement; and
- b. In the event the Consultant's services extend into a future policy period, the Consultant shall, prior to the policy expiration date, provide the City with a new certificate of insurance demonstrating that such coverage is or shall be in effect during all periods of time that Consultant will provide services under this Agreement; and
 - c. Consultant shall maintain said "claims made" coverage for a period of five (5) years following the last date that Consultant has provided services under this Agreement; and
 - d. In the event the Consultant or the insurer terminates "claims made" coverage prior to the expiration of the periods provided in subparagraphs a., b., or c. of this paragraph, the Consultant shall provide to the City advance written notification of the termination of said coverage and shall provide the City with an endorsement for an extended reporting period ("tail coverage") which shall be in effect for a period of time not less than five (5) years following the last date that Consultant has provided services under this Agreement.

Additional Insurance Information

The Consultant shall name the City of Cheyenne as **an Additional Insured** by endorsement on its insurance policies and shall provide the City with a copy of the endorsements. This requirement does not apply to workers' compensation and professional liability insurance policies.

Consultant shall provide the City with certificates of insurance acknowledging the above-stated coverages prior to beginning any work under this Agreement.

It is understood and agreed that these policies are primary and not contributory. All policies required under this Agreement shall be in effect for the duration of the Agreement. It shall be an affirmative obligation upon Consultant to immediately notify in writing the city risk manager, city clerk, and city attorney of any fact, circumstance, or occurrence that has resulted in or may result in the cancellation of, or substantive change to any insurance coverage required by this Agreement, and failure to do so shall be construed to be a breach of this Agreement.

If requested by the City, the Consultant shall provide the City with copies of insurance policies and/or policy endorsements listing the City of Cheyenne as an additional insured. The City's failure to request or review such policies, endorsements, or certificates shall not affect the City's rights or Consultant's obligation hereunder.

Any insurance company providing coverage under this Agreement shall have a minimum A. M. Best rating of A- (excellent).

8. MINORITY AND DISADVANTAGED BUSINESS ENTERPRISES 49 C.F.R. Part 26
All parties to this Agreement assure that no person will be excluded from participation in, denied the benefits of, or otherwise discriminated against, in connection with the award and performance of this Agreement on the grounds of age, race, color, disability, national origin or sex.
9. COMPENSATION
In consideration of the services to be performed pursuant to this Agreement, the Consultant will bill the City and the City agrees to pay Consultant a not to exceed amount of One-Hundred Ninety Nine Thousand Six Hundred one Dollars and Fifty Cents. (\$199,601.50) stated in Exhibit A and Exhibit B.
10. MONTHLY REPORT
With every monthly billing, Consultant shall provide the City Representative with a written statement of the status of the work with respect to the Scope of Work, time sheets, and work schedule. Failure to provide the required monthly report will delay processing of any payment request until the report is submitted.
11. INDEPENDENT CONSULTANT
The Consultant shall function as an independent consultant for the purposes of this Agreement. The Consultant shall assume sole responsibility for any debts or liabilities that may be incurred by the Consultant in fulfilling the terms of this Agreement. Nothing in this Agreement shall be interpreted as authorizing the Consultant or its agents or employees to act as an agent or representative of or on behalf of the City or to incur any obligation of any kind on the behalf of the City.
12. TAXES
The Consultant agrees to pay all valid taxes, excises, license fees, permit fees, bills, debts and obligations incurred by and in connection with its operations under this Agreement.
13. DEFAULT
Each and every term and condition in this Agreement shall be deemed to be a material element of the Agreement. In the event either party should fail or refuse to perform according to the terms of this Agreement, such party may be declared in default.
14. REMEDIES
In the event a party has been declared in default, such defaulting party shall be allowed a period of fifteen (15) days within which to cure the default. In the event the default remains uncorrected, the non-defaulting party declaring default may elect to:
 - a. Terminate the Agreement and seek damages, which damages shall not exceed the contract amount; or
 - b. Treat the Agreement as continuing and require specific performance.

15. TERM AND TERMINATION

This Agreement shall end on or before May 1, 2025. The parties may by mutual written agreement renew or extend this Agreement.

The City may, without cause, and upon thirty days' written notice to the Consultant, terminate this Agreement in whole or in part at any time, for the City's convenience. Upon receipt of such notice, the Consultant shall:

- a. Discontinue all services affected; and
- b. Deliver to the City within five (5) days all data, drawings, specifications, reports, estimates, summaries, and such other information and materials as may have been accumulated by the Consultant in performing this Agreement, whether completed or in process.
- c. In the event of termination for convenience, the City will pay the Consultant for accepted work done and expenses incurred to the date of termination. Such acceptance shall not be unreasonably withheld.

16. ADDITIONAL REMEDIES

In the event the Consultant fails to strictly perform in accordance with this Agreement, the City may elect to correct the deficiencies and charge the Consultant. In the event of default of any of the conditions by either party which shall require the party not in default to commence legal or equitable action against the defaulting party each party shall bear its own costs and expenses, including without limitation, attorney's fees.

17. GOVERNING LAW, JURISDICTION AND VENUE

The construction, interpretation and enforcement of this Agreement shall be governed by the laws of the State of Wyoming. The courts of the State of Wyoming shall have jurisdiction over any action arising out of this Agreement and over the parties, and the venue shall be the First Judicial District, Laramie County, Wyoming.

18. GOVERNMENTAL IMMUNITY

The City of Cheyenne, along with its officials and employees, does not waive its governmental immunity by entering into this Agreement, except to the extent necessary for the parties to pursue a contract action to clarify or enforce the written terms of the Agreement. Furthermore, the City specifically retains all immunities and defenses available to it as a sovereign or governmental entity pursuant to Wyo. Stat. § 1-39-101, *et seq.*, and all other relevant state and federal law. Designations of venue, choice of law, enforcement actions, and similar provisions should not be constructed as a waiver of governmental immunity. The parties agree that any ambiguity in this Agreement shall not be strictly construed, either against or for either party, except that any ambiguity as to governmental or sovereign immunity shall be construed in favor of governmental immunity.

19. MONITORING ACTIVITIES

The City shall have the right to monitor all activities related to this Agreement that are performed by the Consultant or its sub-consultants. This shall include, but not be limited

to, the right to make site inspections at any time and with reasonable notice; to bring experts and consultants on site to examine or evaluate completed work or work in progress; to examine the books, ledgers, documents, papers, and records pertinent to this Agreement; and to observe personnel in every phase of performance of the related work.

20. OWNERSHIP OF DOCUMENTS AND WORK PRODUCT

All deliverables by the Consultant, all work product of the Consultant, all notes, calculations, memoranda, or any other documentation generated by the Consultant pursuant to the scope of this Agreement, shall be the property of the City of Cheyenne and shall be provided to the City of Cheyenne immediately upon request.

21. NON-DISCRIMINATION

The Consultant shall comply with the Civil Rights Act of 1964, the Wyoming Fair Employment Practices Act (Wyo. Stat. § 27-9-105, *et seq.*), the Americans With Disabilities Act (ADA), as amended, 42 U.S.C. § 12101, *et seq.*, and the Age Discrimination Act of 1975 and any properly promulgated rules and regulations thereto, and shall not discriminate against any individual on the grounds of age, sex, creed, color, race, religion, national origin, ancestry, pregnancy or qualifying disability in connection with the performance under this Agreement.

22. PROFESSIONAL REGISTRATION

The Consultant shall endorse, as required by law, plans and reports prepared under this Agreement, and shall affix thereto his or her seal of professional registration, showing that he or she is licensed to practice in the State of Wyoming, if necessary.

23. ENTIRETY OF AGREEMENT

This Agreement consisting of nine (9) pages, and the proposal with Exhibit A and Exhibit B consisting of twenty six (26) pages contain the entire understanding of the parties. There are no other terms or conditions, written or oral, concerning or controlling this matter.

[THIS SPACE INTENTIONALLY LEFT BLANK]

IN WITNESS WHEREOF, the parties have hereunto set their hands and seals the day and year first above written.

City of Cheyenne

(SEAL)

Attest:

Kristina F. Jones, City Clerk

By: _____
Patrick J. Collins, Mayor

Date: _____

Consultant

By: _____

Print Name: _____

Date: _____

SUPPLEMENTAL CONDITIONS TO THE GENERAL CONDITIONS

A. GENERAL CONDITIONS

Standard of Care. The standard of care for all professional engineering and related services performed or furnished by Engineer under this Agreement will be the care and skill ordinarily used by members of the subject profession practicing under similar circumstances at the same time and in the same locality. The Engineer makes no warranties, express or implied, under this Agreement or otherwise, in connection with any services performed or furnished by Engineer.

Technical Accuracy. Owner shall not be responsible for discovering deficiencies in the technical accuracy of Engineer's services. Engineer shall correct deficiencies in technical accuracy without additional compensation unless such corrective action is directly attributable to deficiencies in Owner-furnished information.

Conflicts within the Agreement. If there is a conflict within this Agreement, the order of priority of the documents which make up this Agreement, from highest to lowest, will be:

- (a) Agreement.
- (c) Any supplementary conditions to the General Conditions.
- (d) General Conditions.
- (e) Exhibit A – SCOPE OF SERVICES.
- (f) Other schedules to this Agreement.

Any changes in the Consultant's or Subconsultant's staff or fee structure will be presented in writing to the City for approval prior to initiating any changes or performing any work.

Overhead and Normal Business Expenses. The consultant shall include customary overhead costs and statutory benefits as part including, but not limited to, social security contributions, unemployment, excise and payroll taxes, workers' compensation, health and retirement benefits, bonuses, sick leave, vacation, and holiday pay, as applicable, within the billing rates furnished in Exhibit B. Additionally, the cost of general and administrative overhead which includes salaries and wages of employees engaged in business operations not directly chargeable to projects, plus non-Project operating costs, including but not limited to, business taxes, legal, rent, utilities, office supplies, insurance, and other operating costs shall not be directly billed to project and assumed to be included within the billing rates furnished.

B. DESIGN PROCESS DOCUMENTATION

To document the design process, the consultant will be required to provide a Decision and Design Log for the project duration. This log shall be furnished to the City of Cheyenne in conjunction with the project steering committee meetings. The purpose of the log is to document and track major decisions during the design and development process. The log shall generally follow the format outlined below in Table 1 Design and Design Log:

Table 1 Design and Design Log

No.	Date	Decision	Alternatives Considered	Reasons	Decision Made By	Additional Task Required
Sequenced log number.	Date the decision was made.	A one-sentence summary of the decision made.	A list of the other approaches considered.	A two to three sentence summary of why the decision was made. A link to additional documentation, if required and [Title].	Who made this decision?	If applicable, a list of additional tasks required by the design or follow-up required.

2023



Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne



AVI Professional Corporation
AVI Cheyenne 1103 Old Town Lane , Ste. 101 • Cheyenne, WY 82009
P 307.637.6017 • F 307.632.9326
3/1/2023



Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

Table of Contents

Cover Letter	1
Background & Existing Conditions	2
Location 1	3
Location 2	3
Location 3	3
Location 4A	4
Location 4B.....	4
Location 4C.....	4
Location 4D	5
Exhibit A - Scope of Work	6
Task 1: Project Management	6
Task 2: Research and Reports.....	6
Task 3: Hydraulics and Modeling, Geotechnical (ACE, Terracon, Subconsultants)	7
Task 4: Preliminary Design Phases (30%, 60% & 90%)	7
Task 5: Final Design Phase (100% Construction Documents)	8
Task 6: Maintenance Program & Future Project Development.....	8
Task 7: Bidding and Contract Award	9
Project Team.....	9
Proposed Schedule	10
Exhibit B - Fee Proposal-Compensation	11
Exhibit C – Anderson Consulting Proposal	13
Exhibit D – Proposal Locations 1-7 Exhibit	15

Cover Letter

Thomas Cobb, P.E.
City Engineer
City of Cheyenne, WY
2101 O'Neil Avenue
Cheyenne, WY 82009
VIA EMAIL: tcobb@cheyennecity.org

April 5, 2023
23.011

RE: PROPOSAL FOR DRY CREEK DREDGING & MAINTENANCE PROGRAM PROJECT

Dear Mr. Cobb,

AVI is pleased to submit the following proposal to complete engineering services for the above referenced project. We understand the City of Cheyenne requires professional services to complete the planning, survey, design, and bidding support associated with the proposed Project. Construction management and staking scope and fees can be negotiated following design completion. We also understand that this contract will extend two (2) years, concluding approximately April 2025. We have incorporated two (2) sets of bidding packages within the Scope as requested.

As you are aware, AVI is well positioned to complete this work due to our extensive design history in the area. We believe this background knowledge of the issues related to creek sedimentation will provide for a timely and efficient design the City can utilize to bid and construct the Project.

If you have any questions, please contact our office at your convenience.

Respectfully Submitted

AVI PROFESSIONAL CORPORATION



Tristan Cordier, P.E.
Principal

1103 Old Town Lane | Cheyenne, Wyoming 82009 | 307.637.6017
tcordier@avipc.com avi@avipc.com | www.avipc.com

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Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

BACKGROUND & EXISTING CONDITIONS

Discussions have taken place between City Staff and AVI starting in early fall of 2022. Excessive sedimentation has developed within the Dry Creek drainage basin as it meanders throughout the City. This sedimentation accumulation lessens flood containment and the ability of storm water infrastructure and the creek bed to hydraulically function as designed. Over time, these structures may even become completely silted in, with flooding risk becoming imminent. To lessen this risk, and to remain compliant with federal requirements, urban creeks and infrastructure requires ongoing maintenance which may include debris removal, sedimentation removal, tree, and vegetation removal; back to a known floodway cross section based on hydraulic analysis and historical studies and reporting.

Through discussions with the City and various representatives described below, we understand currently:

- Multiple locations within the basin are in immediate need of clearing and/or dredging.
- The City is currently out of compliance with the Nationwide 404 permit and has been for some time, this is mostly due to lack of revenue to complete maintenance efforts throughout the basin.
- A maintenance program has not been developed for the basin and therefore will be included in the Project scope items discussed in this proposal.
- The City is under contract with West Environmental who is providing services to include a survey and documentation of any species requiring mitigation efforts during the course of the proposed Project. West will also be determining local, state, and federal permitting requirements for each of the locations discussed below.
- The Dry Creek basin is currently under a flood study by GLM Design Group and is being conducted by Mr. Gene McDonald. One task of this study includes determining which sections, structures, etc. within the basin are in the direst need of clearing and/or dredging.
- Additional storm water storage would significantly benefit Location 4A described below and proposed design will incorporate this expansion. Existing concrete pathways will be re-configured and a portion of the existing swale adjacent to Education Dr. will be piped & backfilled.
- Project work should remain within existing easements, rights of way, or property owned by the City.
- Multiple locations described below include heavy vegetation and tree canopies. Tree removals are anticipated.
- A multi-year Project dredging plan is anticipated. Locations 1-7 will be engineered for dredging/clearing and engineer estimates will be provided to the City. Following final design, and document submittal, the City will determine which locations will be best suited for winter 2023/2024, winter 2024/2025 and winter 2025/2026 construction.
- The City of Cheyenne will complete all bidding and advertising. AVI will complete efforts for bidding in a supporting role.

With the guidance of Mr. McDonald and City Staff, the locations below were determined to be included with this Project. For reference, the location area and length, number of plats adjacent, and types of known utilities within the location are highlighted. See **Exhibit D** for Locations 1-7 Exhibit for review.

Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

Location 1

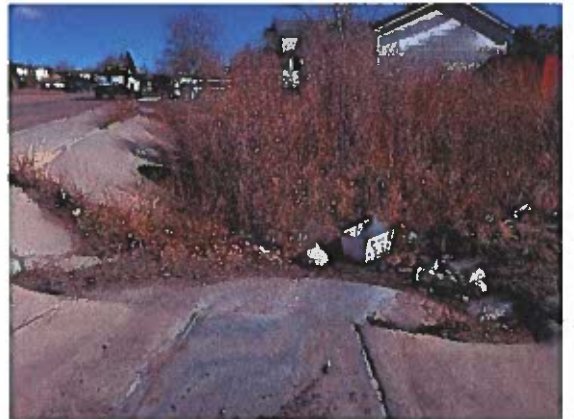
- Dry Creek between Ridge Rd. and N. College Dr.
 - Area = 6.2 Ac.
 - Segment Length = 2,200 LF
 - Number of adjacent Plats = Six
 - Known Utilities
 - Sanitary Sewer
 - Transmission and Distribution Mains
 - Storm Sewer
 - Communications
 - Gas & Electric



Location 1

Location 2

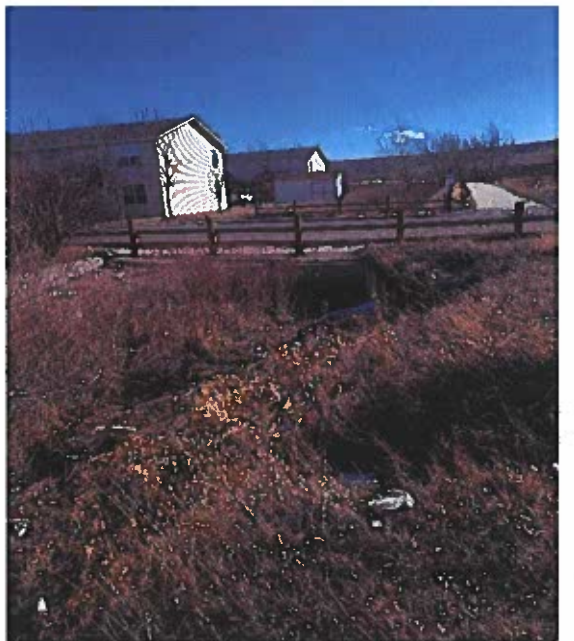
- Holmes Ditch, (Holmes St. south to E. Lincolnway)
 - Area = 1.85 Ac.
 - Segment Length = 1,800 LF
 - Number of adjacent Plats = Four
 - Known Utilities
 - Sanitary Sewer
 - Transmission and Distribution Mains
 - Storm Sewer
 - Communications
 - Gas & Electric



Location 2

Location 3

- Meadowland Ditch (north-south adjacent the Guadalajara Restaurant)
 - Area = 1.82 Ac.
 - Segment Length = 1,200 LF
 - Number of adjacent Plats = Four
 - Known Utilities
 - Sanitary Sewer
 - Transmission and Distribution Mains
 - Storm Sewer
 - Communications
 - Gas & Electric



Location 3

Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

Location 4A

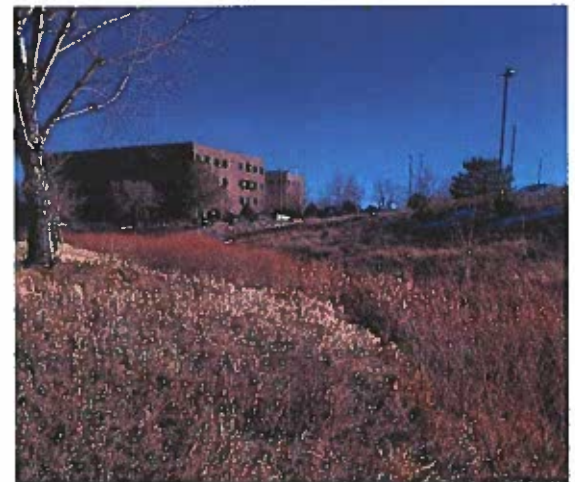
- Dry Creek between McCormick Jr. High School & Central High School
 - Area = 13.4 Ac.
 - Segment Length = 1,700 LF
 - Number of adjacent Plats = Seven
 - Known Utilities
 - Sanitary Sewer
 - Transmission and Distribution Mains
 - Storm Sewer
 - Communications
 - Gas & Electric



Location 4A

Location 4B

- Dry Creek immediately east of Yellowstone Rd.
 - Area = 1.17 Ac.
 - Segment Length = 200 to 300 LF
 - Number of adjacent Plats = Four
 - Known Utilities
 - Sanitary Sewer
 - Transmission and Distribution Mains
 - Storm Sewer
 - Communications
 - Gas & Electric



Location 4B

Location 4C

- Dry Creek, Mylar Park north
 - Area = 2.89 Ac.
 - Segment Length = 700 LF
 - Number of adjacent Plats = Five
 - Known Utilities
 - Sanitary Sewer
 - Transmission and Distribution Mains
 - Storm Sewer
 - Communications
 - Gas & Electric



Location 4C

Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

Location 4D

- Dry Creek, Mylar Park south
 - Area = 3.39 Ac.
 - Creek Length = 600 LF
 - Number of adjacent Plats = Five
 - Known Utilities
 - Sanitary Sewer
 - Transmission and Distribution Mains
 - Storm Sewer
 - Communications
 - Gas & Electric



Location 4D

The quantity of plats noted above are included for reference. We understand the proposed Project intends to remain within the existing easements, roadways or City owned property. However, should a location require any efforts to delineate legal property lines, i.e. due to construction encroachment, plat information will need to be reviewed by one of AVI's registered surveyors to complete any boundary work necessary. Please note that boundary work is not included with any task in this proposal at this time.

As described above, Locations 2 & 3 do not lie within the Dry Creek floodway directly. However, these locations are within the Dry Creek basin, require clearing and/or dredging efforts, and lend themselves to frequent nuisance calls by City residents. Justification for these areas being included within the Project scope are believed to be merited.

Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

EXHIBIT A - SCOPE OF WORK

The Project requires a detailed and phased, or "task" oriented, approach. The tasks are described below and detailed fees following the described items can be seen in **Exhibit B**, "Proposed Fee Schedule Breakdown."

Task 1: Project Management

This portion of services includes general ongoing management and administrative items associated with the Project including stakeholder coordination, internal and external meetings, scheduling, reporting and subconsultant management.

AVI proposes to complete:

- General correspondence, emails, phone communications.
- Subconsultant coordination (ACE, Terracon).
- Project Execution Plan, (PEP). AVI will draft documents associated with the project approach and update during development of the Project.
- Report progress and schedule updates monthly to City Project Manager.
- General project coordination.
- Submit monthly invoicing, with backup documentation as required.
- Facilitate the following meetings:
 - Kickoff meeting with stakeholders.
 - Technical workshop.
 - Eight (8) total monthly progress meetings.
 - One (1) total public meeting.
 - 30% design review with stakeholders.
 - 60% design review with stakeholders.
 - 90% design review with stakeholders.
 - 100% final construction document review with stakeholders.
 - Up to five (5) total external coordination meetings. (i.e. residents, landowners, LCSD#1, BOPU and private utility owners, (ie. Black Hills Energy)).

Task 2: Research and Reports

With this Task, AVI proposes to complete:

- Private utility locates for each location excluding Location 2 & 3 as these will be minor improvements.
- Topographic survey of each location, in State Plane Coordinate System. Survey will include contours, private and public utilities, drainage structures and features, surfacing, and trees 3" diameter or greater and shrub vegetation delineation. Property corners will be surveyed if visible and accessible. Boundary work is not included with this proposal.
- Utility coordination (BOPU & dry utility, Black Hills Energy etc.).
- Site visit of each location with City Project Manager and Forester. At these site visits, AVI proposes to determine which trees and vegetations are to be removed during construction. Each tree for removal will be specifically identified, then re-surveyed. This will provide for more accurate bidding costs from contractors. It will also support a thought through process and assurances to the public during discussions.

Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

- Review of existing mapping and records.
- Up to five (5) non-destructive pothole investigations.

Task 3: Hydraulics and Modeling, Geotechnical (ACE, Terracon, Subconsultants)

- See **Exhibit C**, Anderson Consulting Engineers Inc. Proposal No. WYAVI74.
- Geotechnical investigations (Terracon, on Fee Schedule under "Task 3, Subconsultants")
 - Propose four (4) borings, 10-15' depth each for Location 4A. The data provided from these borings will be utilized to determine the feasibility of expanding storm water storage.
 - Provide soil classifications.
 - Include groundwater elevations (if within the boring depth).
 - Does not include any filtration/percolation testing analysis.

Task 4: Preliminary Design Phases (30%, 60% & 90%)

AVI will develop design documents listed below based on Tasks 1-3. The seven locations detailed previously will be incorporated into a single set of construction drawings and documents. However, phasing will be incorporated to complete multiple bidding processes.

AVI proposes to complete for 30% EPR submittal: (Construction drawings will include existing utility linework)

- Preliminary title sheet.
- Preliminary drainage cross sections.
- Preliminary plan and profile sheets.
- Preliminary storm sewer plan and profiles for Location 4A.
- Preliminary grading plan for Location 4A.
- Preliminary pathway re-alignment for Location 4A.
- Preliminary identification of tree and vegetation removal.
- Preliminary identification of sediment and soil disposal locations.

AVI proposes to complete for both 60% and 90% submittals: (Construction drawings will include existing utility linework, and pipe networks for critical areas and/or realignments)

- Hydraulic analysis memorandum reviews.
- Proposed title sheet and general notes.
- Proposed typical section(s).
- Proposed drainage cross sections.
- Proposed drainage plan and profiles.
- Proposed storm sewer plan and profiles for Location 4A.
- Proposed grading plan for Location 4A.
- Pathway re-alignment for Location 4A.
- Proposed tree and vegetation removal schedule.
- Proposed identification of sediment and soil disposal locations.
- Proposed construction drawing details.
- Proposed summary sheet(s).
- Proposed engineers' construction cost estimates.
- Peer review and quality control.

Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

- Special provisions, specifications.

Task 5: Final Design Phase (100% Construction Documents)

This task will incorporate comments from the 90% submittal and include items associated with bidding support.

AVI will complete:

- Final hydraulic analysis memorandum reviews.
- Final title sheet and general notes.
- Final typical section(s).
- Final drainage cross sections.
- Final drainage plan and profiles.
- Final storm sewer plan & profile sheets for Location 4A.
- Final grading plan for Location 4A.
- Final pathway re-alignment for Location 4A.
- Final tree and vegetation removal schedule.
- Final identification of sediment and soil disposal locations.
- Final construction drawing details.
- Final summary sheet(s); Total estimated quantity.
- Final engineers construction cost estimates.
- Final peer review and quality control.
- Specifications, special conditions, special provisions (Project Manual).
 - Special provisions.
 - Construction specifications.
 - Supplemental specifications.
 - Bid tabulation sheet(s), formatted for multiple bids.
- Provide final information to the public, utility companies, and other required stakeholders.

Task 6: Maintenance Program & Future Project Development

This Task includes efforts to develop a 3-5 year maintenance program and future project schedule for Dry Creek and Crow Creek basins. AVI will coordinate with City staff and stakeholders to review other locations within the basins currently requiring attention. A matrix is proposed to be developed to aid in decision making and order of precedence. This matrix will include inventory of structures in the immediate vicinity of the identified locations, cost estimates for design and survey, and also construction cost estimates based on bidding results for the seven locations identified in this Proposal. AVI assumes that the City will provide a projected annual budget available for these maintenance and future projects. To summarize this Task, AVI will:

- Facilitate meetings to identify up to 6 additional locations.
- Draft location exhibits.
- Inventory adjacent storm sewer infrastructure.
- Develop cost estimates.
- Develop matrices and memorandums of each location.

Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

Task 7: Bidding and Contract Award

This Task shall include support to the City of Cheyenne to complete the bidding process and assist in award of the Project to a qualified contractor with two (2) bidding packages. AVI proposes to complete for the two proposed construction periods (23/24, 24/25,):

- Attend in a supporting role the pre-bid conference.
- Draft addendums and submit to the City for disbursement to bidders.
- Provide to the City, written responses to contractor questions.
- Evaluate bids and make recommendation of award to the City .

The services as outlined above include various meetings, coordination efforts etc. Further details and associated fees can be seen in the attached **Exhibit B**.

PROJECT TEAM

Name	Title	Project Role
<i>Tristan Cordier, PE</i>	Principal	Project Manager
↓		
<i>Bruce Perryman, PE, LS</i>	Sr. Principal	Peer Review
↓		
<i>Blane Cowley</i>	Civil Staff Engineer	Design Drafting
↓		
<i>Adam Deschler, LS</i>	Survey Manager	Surveying
↓		
<i>Craig Cowley</i>	Crew Chief	Surveying
↓		
<i>Jamis Darrow, PE, CFM</i>	Anderson Consulting Engineers, Inc.	Hydraulic Analysis
↓		
<i>Zachary St. Jean, PE, PG</i>	Terracon Consultant	GEO Tech

Resumes are available upon request.

Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

PROPOSED SCHEDULE

Subject to adjustment pending final contract executions, weather, and scope changes presented during kickoff discussions with City Staff.

Date	Schedule Item
Mid April 2023	Notice-to-Proceed issued by City
April 20 th , 2023	Kickoff Meeting
April 17 th – May 26 th , 2023	Task 2 Research and Reports
May 29 th -	Task 3 Hydraulics & Modeling
May 1 st – June 16 th , 2023	Task 3 Geotechnical, Borings and Reports
May 1 st – August 26 th , 2023	Task 4 Preliminary Design
September 13 th	Public Meeting
September 4 th – September 29 th , 2023	Task 5 Final Design
Ongoing through 2023	Task 6 Maintenance and Program Development
November 2023	Task 7 Bidding & Contract Award One
November 2024	Task 7 Bidding & Contract Award Two

Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

EXHIBIT B - FEE PROPOSAL-COMPENSATION

AVI's not-to-exceed fee for this work is **\$199,601.50**. Once you have reviewed the above scope of services, we would be most interested in assisting you with the successful completion of this project. See **Exhibit B** for Proposed Fee Schedule Breakdown.

AVI FEE PROPOSAL SUMMARY for Dry Creek Dredging & Maintenance Project

Task	Description	\$175.00		\$140.00		\$104.00		\$70.00		\$35.00		\$140.00		\$112.00		Estimated Sub-total
		Senior Principal Peer Review & QC		Sr. Registered Engineer		Engineer (Non Registered)		Bookkeeping		Clerical		Registered Surveyor		Party Chief		
		Hrs	Cost	Hrs	Cost	Hrs	Cost	Hrs	Cost	Hrs	Cost	Hrs	Cost	Hrs	Cost	
Task 1 - Project Management																
a	General Correspondence and Communication	2	\$ 350.00	40	\$ 5,600.00		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 5,950.00
b	Subconsultant Coordination		\$ -	24	\$ 3,360.00		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 3,360.00
c	Project Evaluation Plan (PEP)		\$ -	8	\$ 1,120.00		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 1,120.00
d	Monthly Progress Reports and Schedule Updates		\$ -	16	\$ 2,240.00		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 2,240.00
e	General Project Coordination	8	\$ 1,400.00	24	\$ 3,360.00	12	\$ 1,680.00		\$ -		\$ -		\$ -		\$ -	\$ 6,012.00
f	Monthly Invoicing		\$ -	8	\$ 1,120.00		\$ -	16	\$ 2,240.00		\$ -		\$ -		\$ -	\$ 3,360.00
g	Kickoff Meeting	2	\$ 350.00	2	\$ 280.00	2	\$ 208.00		\$ -		\$ -		\$ -		\$ -	\$ 840.00
h	Technical Workshop	2	\$ 350.00	2	\$ 280.00	2	\$ 208.00		\$ -		\$ -		\$ -		\$ -	\$ 840.00
i	Monthly Progress Meeting (if Total)	8	\$ 1,400.00	8	\$ 1,120.00	8	\$ 832.00		\$ -		\$ -		\$ -		\$ -	\$ 3,360.00
j	Public Meeting (1 Total)	4	\$ 700.00	4	\$ 560.00	4	\$ 416.00		\$ -		\$ -		\$ -		\$ -	\$ 1,680.00
k	Design Review Meetings	4	\$ 700.00	4	\$ 560.00		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 1,260.00
l	External Coordination Meeting (5 Total)	3	\$ 525.00	6	\$ 840.00		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 1,371.00
m	Miscellaneous Expenses & Consumables		\$ -		\$ -		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 500.00
Task 1 Subtotal		33	\$ 5,750.00	148	\$ 20,560.00	28	\$ 3,816.00	16	\$ 2,240.00		\$ -		\$ -		\$ -	\$ 29,893.00
Task 2 - Research & Reports																
a	Private Utility Locates		\$ -	4	\$ 560.00		\$ -		\$ -		\$ -	28	\$ 3,920.00	185	\$ 21,460.00	\$ 25,559.00
b	Topographic Survey	1	\$ 175.00		\$ -		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 175.00
c	Utility Coordination (BOP/Utility)	4	\$ 700.00	16	\$ 2,240.00		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 2,940.00
d	Site Visit		\$ -	14	\$ 1,974.00		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 1,974.00
e	Review Existing Mapping & Records		\$ -	2	\$ 280.00		\$ -		\$ -		\$ -	10	\$ 1,400.00	8	\$ 928.00	\$ 2,608.00
f	Utility Profile Excavations - Non Destructive (5 Total)		\$ -		\$ -		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 500.00
g	Miscellaneous Expenses & Consumables		\$ -		\$ -		\$ -		\$ -		\$ -	14	\$ 1,974.00	193	\$ 22,388.00	\$ 24,862.00
Task 2 Subtotal		5	\$ 875.00	36	\$ 5,084.00		\$ -		\$ -		\$ -	34	\$ 4,760.00	193	\$ 22,388.00	\$ 30,615.00
Task 3 - Construction Design (30%, 60%, 90%)																
30%																
a	Title Sheet		\$ -	0.5	\$ 70.50	1	\$ 104.00		\$ -		\$ -		\$ -		\$ -	\$ 174.50
b	Cross Sections		\$ -	2	\$ 282.00	20	\$ 2,080.00		\$ -		\$ -		\$ -		\$ -	\$ 2,362.00
c	Plan & Profiles		\$ -	9	\$ 1,260.00	20	\$ 2,080.00		\$ -		\$ -		\$ -		\$ -	\$ 3,340.00
d	Storm Sewer Plan & Profile (Location 4A)		\$ -	3	\$ 416.00	8	\$ 832.00		\$ -		\$ -		\$ -		\$ -	\$ 1,248.00
e	Grading Plans (Location 4A)		\$ -	4	\$ 560.00	30	\$ 3,120.00		\$ -		\$ -		\$ -		\$ -	\$ 3,680.00
f	Highway Re-Alignment (Location 4A)		\$ -	2	\$ 282.00	16	\$ 1,664.00		\$ -		\$ -		\$ -		\$ -	\$ 1,946.00
g	Tree & Vegetation Removals		\$ -	2	\$ 282.00	16	\$ 1,664.00		\$ -		\$ -		\$ -		\$ -	\$ 1,946.00
h	Sedimentation & Disposal Location Recommendations	2	\$ 282.00	6	\$ 846.00	2	\$ 208.00		\$ -		\$ -		\$ -		\$ -	\$ 1,336.00
60%																
a	Hydraulic Analysis Memorandums (Reviews)	2	\$ 282.00	2	\$ 282.00		\$ -		\$ -		\$ -		\$ -		\$ -	\$ 564.00
b	Title Sheet		\$ -	0.5	\$ 70.50	0.5	\$ 52.00		\$ -		\$ -		\$ -		\$ -	\$ 122.50
c	General Notes		\$ -	0.5	\$ 70.50	1	\$ 104.00		\$ -		\$ -		\$ -		\$ -	\$ 174.50
d	Typical Sections		\$ -	4	\$ 560.00	16	\$ 1,664.00		\$ -		\$ -		\$ -		\$ -	\$ 2,224.00
e	Cross Sections		\$ -	4	\$ 560.00	32	\$ 3,328.00		\$ -		\$ -		\$ -		\$ -	\$ 3,888.00
f	Plan & Profiles		\$ -	2	\$ 282.00	32	\$ 3,328.00		\$ -		\$ -		\$ -		\$ -	\$ 3,610.00
g	Storm Sewer Plan & Profile (Location 4A)		\$ -	2	\$ 282.00	8	\$ 832.00		\$ -		\$ -		\$ -		\$ -	\$ 1,114.00
h	Grading Plans (Location 4A)		\$ -	4	\$ 560.00	16	\$ 1,664.00		\$ -		\$ -		\$ -		\$ -	\$ 2,224.00

L	Pathway Re Alignment (Location 4A)	\$	2	\$	282.00	12	\$	1,248.00	\$		\$		\$		\$		\$	1,530.00
J	Tree & Vegetation Removals	\$	2	\$	282.00	4	\$	416.00	\$		\$		\$		\$		\$	696.00
K	Sedimentation & Disposal Location Recommendations	\$	2	\$	282.00	1	\$	104.00	\$		\$		\$		\$		\$	386.00
L	Construction Drawing Details	\$	1	\$	141.00	4	\$	416.00	\$		\$		\$		\$		\$	557.00
M	Summary Sheets	\$	2	\$	282.00	4	\$	416.00	\$		\$		\$		\$		\$	696.00
N	Engineers Construction Estimate	\$	2	\$	282.00	16	\$	1,664.00	\$		\$		\$		\$		\$	1,946.00
O	Peer Review & Quality Control	6	\$	1,400.00			\$		\$		\$		\$		\$		\$	1,400.00
P	Special Provisions & Specifications	0.5	\$	87.50	1	\$	141.00	1	\$	104.00	\$		8	\$	560.00	\$		892.50
96%																		
A	Hydraulic Analysis Memorandums (Reviews)	2	\$	350.00	1	\$	141.00		\$		\$		\$		\$		\$	491.00
B	Title Sheet		\$		0.5	\$	70.50	0.5	\$	52.00	\$			\$		\$		122.50
C	General Notes		\$		0.5	\$	70.50	1	\$	104.00	\$			\$		\$		174.50
D	Typical Sections		\$		2	\$	282.00	8	\$	832.00	\$			\$		\$		1,114.00
E	Cross Sections		\$		4	\$	564.00	12	\$	1,248.00	\$			\$		\$		1,812.00
F	Plan & Profiles		\$		4	\$	564.00	8	\$	832.00	\$			\$		\$		1,396.00
G	Storm Sewer Plan & Profile (Location 4A)		\$		2	\$	282.00	6	\$	624.00	\$			\$		\$		906.00
H	Graffiti Plans (Location 4A)		\$		4	\$	564.00	8	\$	832.00	\$			\$		\$		1,396.00
I	Pathway Re Alignment (Location 4A)		\$		2	\$	282.00	8	\$	832.00	\$			\$		\$		1,114.00
J	Tree & Vegetation Removals		\$		1	\$	141.00	1	\$	104.00	\$			\$		\$		245.00
K	Sedimentation & Disposal Location Recommendations		\$		1	\$	141.00	1	\$	104.00	\$			\$		\$		245.00
L	Construction Drawing Details		\$		1	\$	141.00	2	\$	208.00	\$			\$		\$		349.00
M	Summary Sheets		\$		1	\$	141.00	2	\$	208.00	\$			\$		\$		349.00
N	Engineers Construction Estimate		\$		2	\$	282.00	6	\$	832.00	\$			\$		\$		1,114.00
O	Peer Review & Quality Control	4	\$	700.00			\$			\$				\$		\$		700.00
P	Special Provisions & Specifications	0.5	\$	87.50	2	\$	282.00	0.5	\$	52.00	\$		4	\$	280.00	\$		701.50
Q	Miscellaneous Expenses & Consumables										\$					\$		1,000.00
		19	\$	3,325.00	77.5	\$	10,927.50	346.5	\$	36,018.00	\$		12	\$	840.00	\$		52,128.50
A	Hydraulic Analysis Memorandums (Reviews)	1	\$	175.00	1	\$	141.00		\$		\$		\$		\$		\$	316.00
B	Title Sheet		\$		0.5	\$	70.50	0.5	\$	52.00	\$			\$		\$		122.50
C	General Notes		\$		0.5	\$	70.50	0.5	\$	52.00	\$			\$		\$		122.50
D	Typical Sections		\$		2	\$	282.00	2	\$	208.00	\$			\$		\$		490.00
E	Cross Sections		\$		2	\$	282.00	8	\$	832.00	\$			\$		\$		1,114.00
F	Plan & Profiles		\$		2	\$	282.00	8	\$	832.00	\$			\$		\$		1,114.00
G	Storm Sewer Plan & Profile (Location 4A)		\$		1	\$	141.00	2	\$	208.00	\$			\$		\$		349.00
H	Graffiti Plans (Location 4A)	1	\$	175.00	2	\$	282.00	8	\$	832.00	\$			\$		\$		1,289.00
I	Pathway Re Alignment (Location 4A)	1	\$	175.00	2	\$	282.00	8	\$	832.00	\$			\$		\$		1,289.00
J	Tree & Vegetation Removals		\$		1	\$	141.00	0.5	\$	52.00	\$			\$		\$		193.00
K	Sedimentation & Disposal Location Recommendations		\$		1	\$	141.00	0.5	\$	52.00	\$			\$		\$		193.00
L	Construction Drawing Details		\$		1	\$	141.00	2	\$	208.00	\$			\$		\$		349.00
M	Summary Sheets		\$		1	\$	141.00	2	\$	208.00	\$			\$		\$		349.00
N	Engineers Construction Estimate		\$		6	\$	846.00	4	\$	416.00	\$			\$		\$		1,262.00
O	Peer Review & Quality Control	4	\$	700.00			\$			\$				\$		\$		700.00
P	Special Provisions & Specifications	2	\$	350.00	2	\$	282.00	0.5	\$	52.00	\$		4	\$	280.00	\$		964.00
Q	Bid Tabulation Sheet	1	\$	175.00	2	\$	282.00	2	\$	208.00	\$		2	\$	140.00	\$		805.00
R	Miscellaneous Expenses & Consumables										\$					\$		500.00
		10	\$	1,750.00	27	\$	3,807.00	48.5	\$	5,044.00	\$		6	\$	420.00	\$		11,321.00
A	Facilitate Coordination Meetings	4	\$	700.00	4	\$	564.00	4	\$	416.00	\$		\$		\$		\$	1,680.00
B	Draft Location Exhibits	6	\$	1,050.00	2	\$	282.00	12	\$	1,248.00	\$		\$		\$		\$	2,540.00
C	Inventory Adjacent Storm Sewer Infrastructure		\$		4	\$	564.00	8	\$	832.00	\$			\$		\$		1,396.00
D	Develop Cost Estimates	4	\$	700.00	4	\$	564.00	12	\$	1,248.00	\$			\$		\$		2,512.00
E	Develop Mapfiles and Memorandums	4	\$	700.00	4	\$	564.00	12	\$	1,248.00	\$			\$		\$		2,512.00
F	Miscellaneous Expenses & Consumables										\$					\$		500.00
		18	\$	3,150.00	18	\$	2,538.00	48	\$	4,992.00	\$		\$		\$		\$	11,180.00
A	Attend Pre Bid Conference		\$		4	\$	564.00	4	\$	416.00	\$		\$		\$		\$	980.00

[illegible]

AVI PROFESSIONAL CORPORATION

2023 Wyoming Billing Rates



	2023 Rate
Senior Principal	\$175.00 per hour
Principal	\$153.00 per hour
Senior Registered Engineer	\$141.00 per hour
Registered Engineer	\$123.00 per hour
Engineer (Non-Registered)	\$104.00 per hour
Senior Planner	\$141.00 per hour
Drainage Engineer	\$135.00 per hour
Planner	\$104.00 per hour
Construction Manager	\$110.00 per hour
Technician I	\$105.00 per hour
Technician II	\$101.00 per hour
Technician III	\$95.00 per hour
Inspector I	\$95.00 per hour
Bookkeeping	\$70.00 per hour
Clerical	\$70.00 per hour
Registered Surveyor	\$140.00 per hour
Party Chief	\$116.00 per hour
Crew Member	\$65.00 per hour
Party Chief/RLS	\$140.00 per hour

**** ANY MULTI-YEAR PROJECTS WILL INCLUDE A STANDARD 2% - 3% INCREASE IN RATES FOR EACH YEAR FOLLOWING PROJECT INCEPTION DATE.**

**** Expert witness and/or legal assistance will bill at 2.5 times the above billing rates.**

AVI PROFESSIONAL CORPORATION

2023 Wyoming Billing Rates



2023 Rate

TRAVEL CHARGES

Mileage (Private)	\$0.625 per mile
Mileage (State/Municipality/County)	IRS Rate per mile
Mileage for AVI Staff	\$0.625 per mile
ATV/Boat Rentals	\$150.00 per day
UTV Rental	\$150.00 per day

EQUIPMENT CHARGES

Trimble R6 GPS Survey Equipment	\$320.00 per day per unit
Additional Rover	\$20.00 per hour
Trimble R6 GPS Survey Equipment w/2 nd Rover	\$60.00 per hour
Digital Level (Trimble/Leica)	\$100.00 per day
Robotic Total Station	\$20.00 per hour/\$160.00 per day
Robotic Total Station w/scanner	\$30.00 per hour/\$250.00 per day
Trailer	\$25.00 per day

LARGE FORMAT PRINTS & PLOTS

Copies	\$0.25 per square foot
Mylars	\$2.50 per square foot
Outsourced Copies & Postage	Cost
Office Fees (Copies, Phone, Postage, Etc.)	\$300.00 per month

PERDIEM

Meals	\$40.00/ 8 hour work day
Lodging – If Provided	\$35.00 per day
Lodging – If Not Provided	Actual cost

Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

EXHIBIT C – ANDERSON CONSULTING PROPOSAL



ANDERSON CONSULTING ENGINEERS, INC.
Civil • Water Resources • Environmental

February 28, 2023

Mr. Tristan Cordier
AVI P.C.
1103 Old Town Lane, Ste. 101
Cheyenne, WY 82009

RE: Scope and Budget for Hydraulic Analyses of the Dry Creek Dredging Project
(ACE Project No. WYAVI74)

Dear Tristan:

As requested, attached is a scope, and budget for hydraulic analyses associated with the Dry Creek Dredging project to be completed in Cheyenne, Wyoming. In general, the proposed services include modifying effective models utilizing current topography to create existing condition models, followed by creation of proposed condition models utilizing design plans prepared by AVI. Brief memorandums will be prepared for each site to document whether or not the improvements satisfy no-rise criteria. We propose to complete the work outlined in the accompanying scope for time and materials budget not to exceed of \$32,715.

We respectfully submit this scope and budget based on our understanding of your needs for permitting assistance. After review of this information please do not hesitate to contact us if you have any questions or comments.

Sincerely,

Anderson Consulting Engineers, Inc.

Jamis Darrow, P.E., CFM
Project Manager

Attachments: A. Anderson Consulting Engineers Scope of Work
B. Budget for the Rivers Edge Bank Stabilization Project

EXHIBIT A
SCOPE OF WORK
for the
HYDRAULIC ANALYSES FOR THE DRY CREEK DREDGING PROJECT

At the request of AVI P.C., this scope of work and budget has been developed by Anderson Consulting Engineers, Inc. (ACE) to provide the following engineering services:

- (a) Collection and review of effective hydraulic models;
- (b) Utilization of topography to be collected by AVI to modify effective models to create five existing condition models;
- (c) Modification of the five existing condition models utilizing design drawings prepared by AVI to develop proposed condition models;
- (d) Preparation of brief memorandums for each site to document findings of the hydraulic analyses; and
- (e) Project coordination including general coordination and meetings with the Cheyenne floodplain administrator and City staff.

Assumptions:

- 1. Effective models will be provided by the City of Cheyenne covering each of the five dredging sites. A discussion of the effective models is provided in Task 2.
 - 2. Existing condition survey will be provided by AVI P.C. for each of the five design sites.
 - 3. Design drawings for each site, including design contours and any changes in channel/overbank roughness/vegetative cover, will be prepared by AVI P.C. and submitted to ACE for use in proposed condition modeling.
 - 4. Existing condition survey and proposed contours will be provided in a known coordinate system (i.e. Wyoming State Plane, etc.).
 - 5. Existing condition survey will cover floodplain extents, or supplemental data will be available for model modifications.
 - 6. Project designs will not include any fill in effective floodways.
 - 7. No FEMA submittals will be required.
 - 8. All environmental permitting will be completed by others.
 - 9. Documentation provided by ACE will be utilized by AVI to complete any necessary floodplain use permits.
 - 10. No as-built analyses will be required under this scope and budget.
 - 11. Designs for the five dredging projects will result in no-rise. If a rise is observed, ACE staff will work with AVI under a separate scope and budget to find a solution that results in no-rise.
 - 12. Assuming projects result in no-rise scenarios, no floodplain/floodway mapping will be required.
 - 13. Of the seven sites identified by AVI, hydraulic analyses will not be required for the Meadowland Ditch and Holmes Ditch sites.
-

TASK 1. COLLECTION OF EFFECTIVE DATA

This task will include familiarization of project staff with site hydraulics through the acquisition and review of effective models, and effective floodplain/floodway mapping. Model versions of effective models will be identified so that the same models may be utilized for hydraulic analyses. In addition vertical datum conversions will be identified/documentated to ensure that all mapping/modeling data are consistent.

TASK 2. HYDRAULIC ANALYSES OF DREDGING PROJECTS

For each of the five sites identified by AVI, hydraulic analyses will be conducted to determine existing and proposed condition water surface elevations throughout project sites for the 100-year event. Task 2 has been subdivided into tasks 2.1 through 2.5 as follows:

- Task 2.1: Dry Creek Area 1
- Task 2.2: Dry Creek Area 4A
- Task 2.3: Dry Creek Area 4B
- Task 2.4: Dry Creek Area 4C
- Task 2.5: Dry Creek Area 4D

For each site hydraulic analyses for the specified dredging project will require completion of the following items:

1. Preparation of a project base map;
2. Existing condition hydraulic analyses;
3. Proposed condition hydraulic analyses;
4. Preparation of water surface comparisons; and
5. Reporting

BASE MAP PREPARATION

AVI surveyors will collect field survey data and create topographic site mapping, which reflects existing conditions and will be used to prepare a project base map. The survey will primarily include a detailed channel definition for the project area. It is assumed that site survey will be conducted under a separate scope and budget through AVI and the City of Cheyenne. This site survey will be collected and utilized as part of the project base map.

Design topography prepared by AVI will also be added to the project base map to illustrate differences between existing and proposed conditions.

Finally, effective flood hazard delineations obtained from the City of Cheyenne/FEMA will be added to the project base map to show where project extents exist in relation to the effective floodplain boundary.

HYDRAULIC ANALYSES

Following preparation of the site base map, it will be necessary to ensure that the project design satisfies a “no-rise” criteria to avoid CLOMR/LOMR submittals. Therefore, as part of the design process ACE staff will verify that design conditions for the five dredging projects show no-rise in 100-year flood elevations relative to effective conditions. ACE staff will coordinate with the local floodplain administrator to ensure that the no-rise documentation utilizing model modifications will satisfy local floodplain regulations. The

no-rise analysis will be documented and presented as support documentation for Floodplain Development Permits to be prepared by AVI.

Upon inspection of the effective data provided on the FEMA Map Service Center, it appears that the five project sites are modeled as part of three effective studies. Dry Creek Areas 4A, 4B, 4C, and 4D were all included in the original Dry Creek study, which appears to have been studied using HEC2. Dry Creek Area 1 was included in LOMR # 10-08-0553P, while the upstream portion of Dry Creek Area 4A was studied in LOMR # 17-08-1565P. The version of hydraulic model utilized for these two studies is unknown, but due to the dates these studies became effective it is assumed that they were completed utilizing HEC-RAS.

First, for each site the effective model will be modified to reflect existing conditions. Model modifications will be primarily limited to updating effective cross sections with existing condition topography and adding a limited number of additional cross sections to properly model the project area.

Second, for each site the existing condition model will be modified utilizing design information prepared by AVI to create a proposed condition model. Modifications made for the proposed condition model will be limited to updating site cross sections with proposed topography, and possibly updating roughness coefficients if warranted by design drawings.

It is assumed that all model changes will be conducted utilizing the model version utilized in the effective study. It is also assumed that no corrected effective changes will be required for the effective models (i.e. structure modeling is correct and accurate, no divided flow paths will be required, etc.).

Following hydraulic modeling efforts, tables comparing effective, existing, and proposed condition 100-year water surface elevations will be prepared.

PROJECT REPORTS

A brief summary memorandum will be prepared for each site that summarizes analytical assumptions and methodologies, hydraulic analyses, and analytic results. Pertinent documentation will accompany the report, including electronic copies of the hydraulic models.

TASK 3. MEETINGS AND COORDINATION

This task includes limited project coordination, both in the form of progress meetings to be held via conference call, and one meeting with City Staff to be held via virtual platform (i.e. Zoom, Microsoft Teams, etc.). ACE staff assumes that the project coordination for this project will include the following:

- General project coordination to discuss design intent and project progress; and
- A pre-project meeting with the City's Floodplain Administrator to discuss project approach;
- One meeting with City staff to discuss project results.

EXHIBIT B

Hydraulic Analyses for the Dry Creek Dredging Project

Task	Anderson Consulting				Other Direct Costs	Sub-Task Total
	Senior Principal Engineer	Project Manager	Project Engineer I	Senior GIS Technician/ Drone Pilot		
	\$210	\$150	\$105	\$105		
Task 1: Project Background						\$2,070
Collection/Review of existing materials	1	4	9	3	\$0	\$2,070
Task 2.1: Dry Creek Area 1 Analysis/Reporting						\$5,430
Base Map Preparation		1	5	2	\$0	\$885
Existing Condition Modeling		1	14		\$0	\$1,620
Proposed Condition Modeling		1	9		\$0	\$1,095
Water Surface Comparison Tables			3		\$0	\$315
Technical Memorandum		8	2	1	\$0	\$1,515
Task 2.2: Dry Creek Area 4A Analysis/Reporting						\$7,425
Base Map Preparation		1	5	2	\$0	\$885
Existing Condition Modeling		1	26		\$0	\$2,880
Proposed Condition Modeling		1	15		\$0	\$1,725
Water Surface Comparison Tables			4		\$0	\$420
Technical Memorandum		8	2	1	\$0	\$1,515
Task 2.3: Dry Creek Area 4B Analysis/Reporting						\$4,905
Base Map Preparation		1	5	2	\$0	\$885
Existing Condition Modeling		1	10		\$0	\$1,200
Proposed Condition Modeling		1	9		\$0	\$1,095
Water Surface Comparison Tables			2		\$0	\$210
Technical Memorandum		8	2	1	\$0	\$1,515
Task 2.4: Dry Creek Area 4C Analysis/Reporting						\$5,220
Base Map Preparation		1	5	2	\$0	\$885
Existing Condition Modeling		1	12		\$0	\$1,410
Proposed Condition Modeling		1	9		\$0	\$1,095
Water Surface Comparison Tables			3		\$0	\$315
Technical Memorandum		8	2	1	\$0	\$1,515
Task 2.5: Dry Creek Area 4D Analysis/Reporting						\$5,220
Base Map Preparation		1	5	2	\$0	\$885
Existing Condition Modeling		1	12		\$0	\$1,410
Proposed Condition Modeling		1	9		\$0	\$1,095
Water Surface Comparison Tables			3		\$0	\$315
Technical Memorandum		8	2	1	\$0	\$1,515
Task 3: Project Coordination						\$2,445
General Project Coordination	1	6	4		\$0	\$1,530
Meeting with Floodplain Administrator		2	1		\$0	\$405
Project Meeting		2	2		\$0	\$510
Labor Category Total Hours	2	69	191	18	0	
Labor Category Total Cost	\$420	\$10,350	\$20,055	\$1,890	\$0	
Project Total						\$32,715

Dry Creek Dredging & Maintenance Project

Proposal For The City of Cheyenne

AVI Professional Corporation

EXHIBIT D – PROPOSAL LOCATIONS 1-7 EXHIBIT

2023 CITY OF CHEYENNE DRAINAGE DREDGING PROJECT

